PRACTICAL 11-B

Object Oriented Programming

AIM:

To add object persistence and reporting by using serialization, file I/O, transaction logs and NIO directory traversal.

CODE SCREENSHOT:

MiniBank.java

Part B > Practical 11-B > MiniBank.java > ...

```java
import java.io.*;
import java.nio.file.*;
import java.util.*;

class Account {
    final String id, owner;
    long balance;

    Account(String i, String o, long b) {
        id = i;
        owner = o;
        balance = b;
    }

    String row() {
        return id + "," + owner + "," + balance;
    }

    static Account parse(String s) {
        String[] p = s.split(",");
        return new Account(p[0], p[1], Long.parseLong(p[2]));
    }

    public String toString() {
        return row();
    }
}

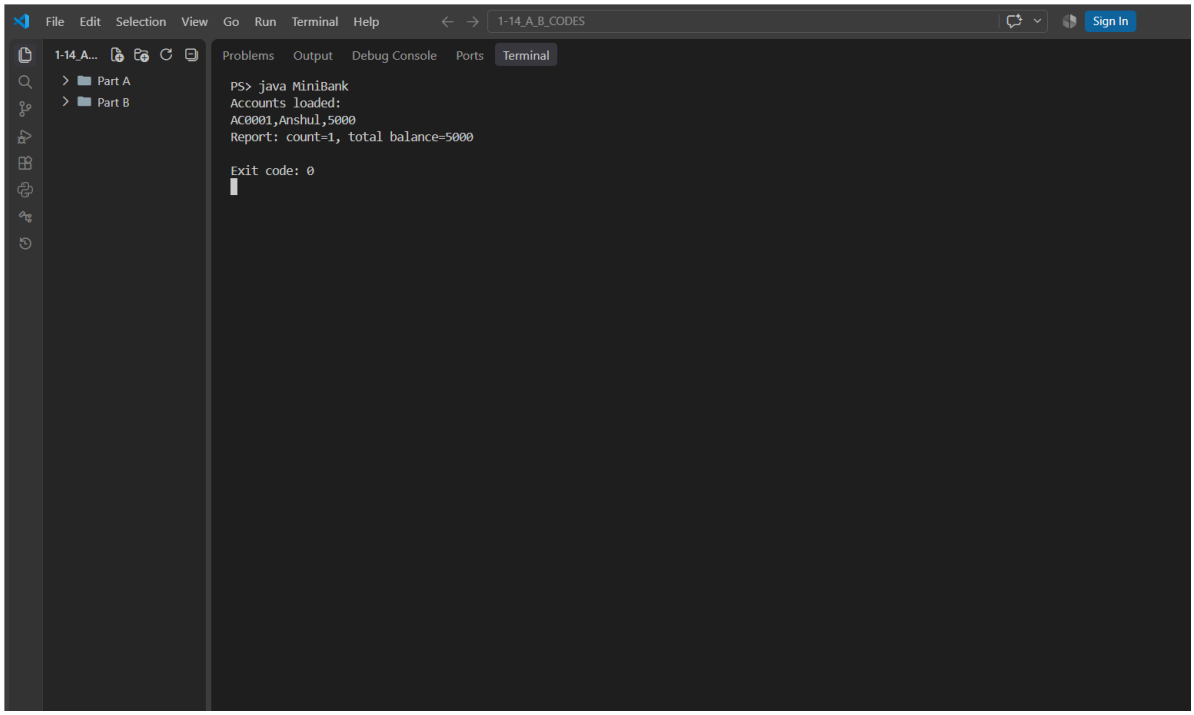
class AccountStore {
    private final Path path = Path.of("minibank-accounts.csv");

    void save(Collection<Account> accounts) throws IOException {
        List<String> lines = new ArrayList<>();
        for (Account a : accounts)
            lines.add(a.row());
        Files.write(path, lines);
    }

    List<Account> load() throws IOException {
        List<Account> out = new ArrayList<>();
        if (Files.exists(path))
            for (String s : Files.readAllLines(path))
                if (!s.isBlank())
                    out.add(Account.parse(s));
        return out;
    }
}

```

OUTPUT SCREENSHOT:



LEARNING OUTCOMES:

The student can persist objects to files, reload them, and summarise transaction logs using NIO.